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OSTİM TECHNICAL UNIVERSITY FACULTY OF ENGINEERING
SOFTWARE ENGINEERING DEPARTMENT

Real-Time Anti-Money Laundering Detection System

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1. PROJECT OBJECTIVE

Design and implement a fully serverless, real-time AML detection pipeline using AWS managed services.

- Real-time transaction ingestion
- Stateful AML detection
- Serverless and scalable architecture
- Analytics layer with AWS Athena
- Interactive dashboard for compliance analysts
- Low latency and high reliability

2. AML Pattern

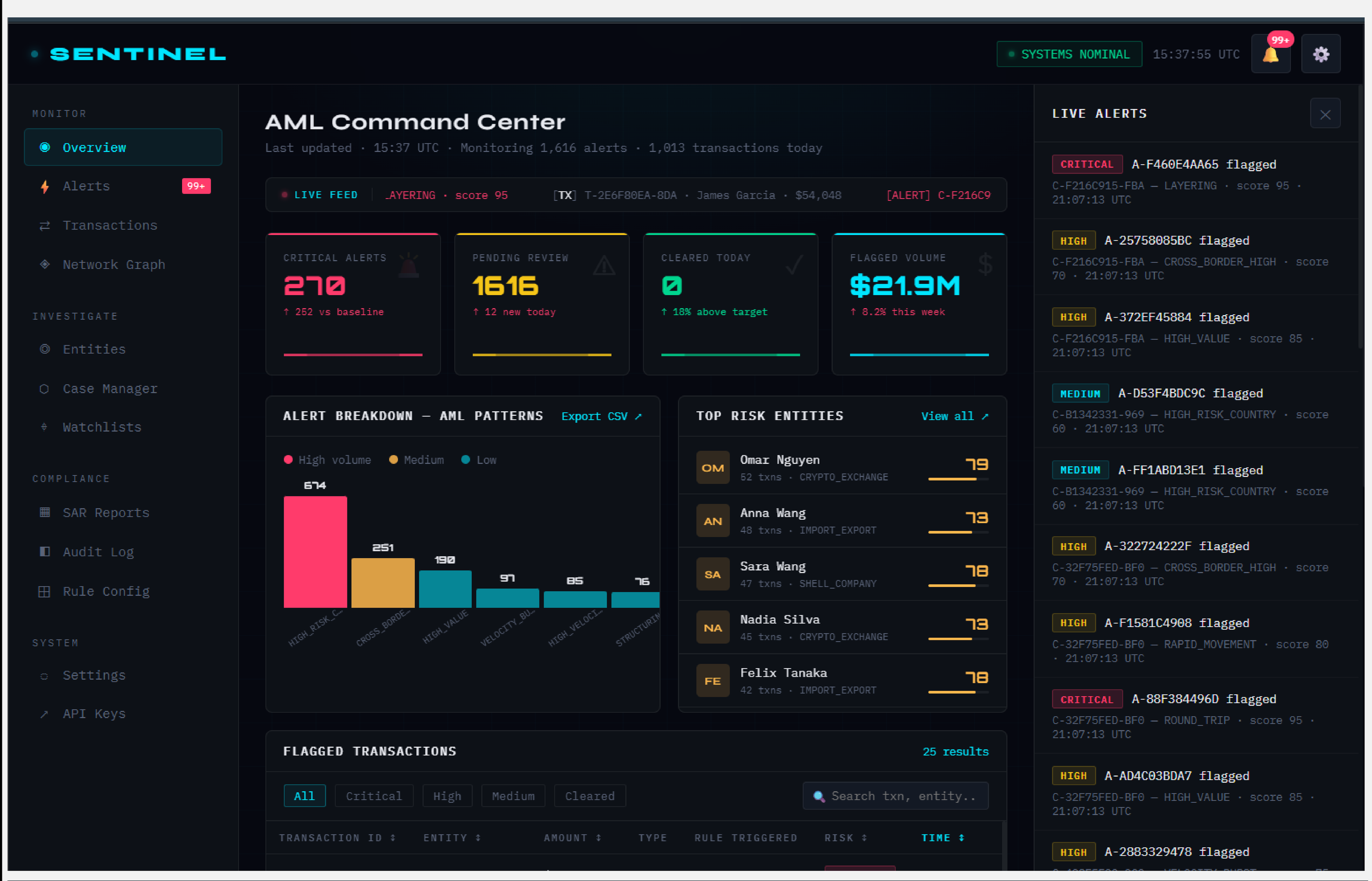
14 Rule-based detection patterns implemented

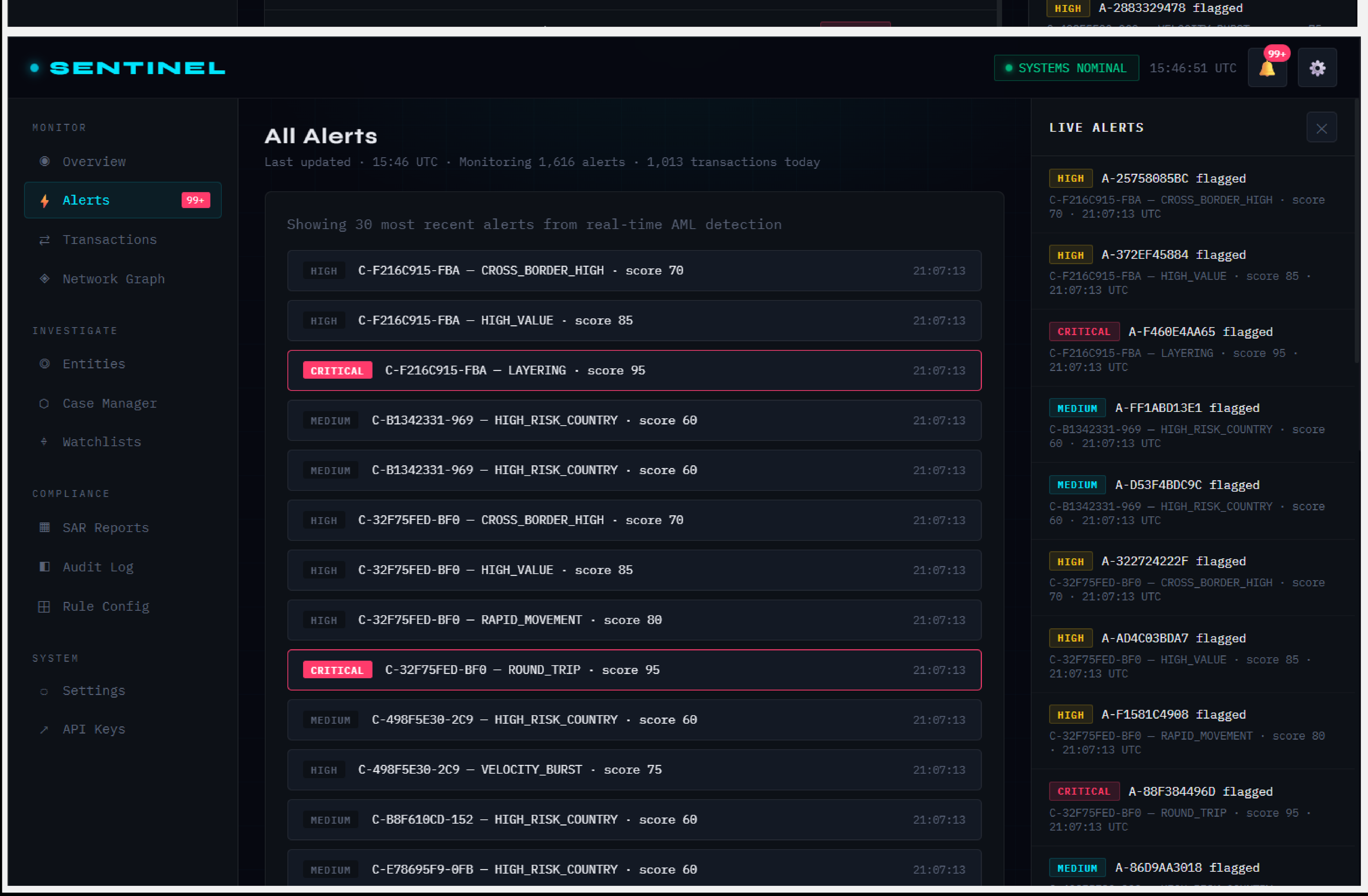
- Stateless Rules
- Stateful Rules

stateless rules protect against individual suspicious *transactions*, while stateful rules protect against suspicious behavior and behavior only reveals itself across time.

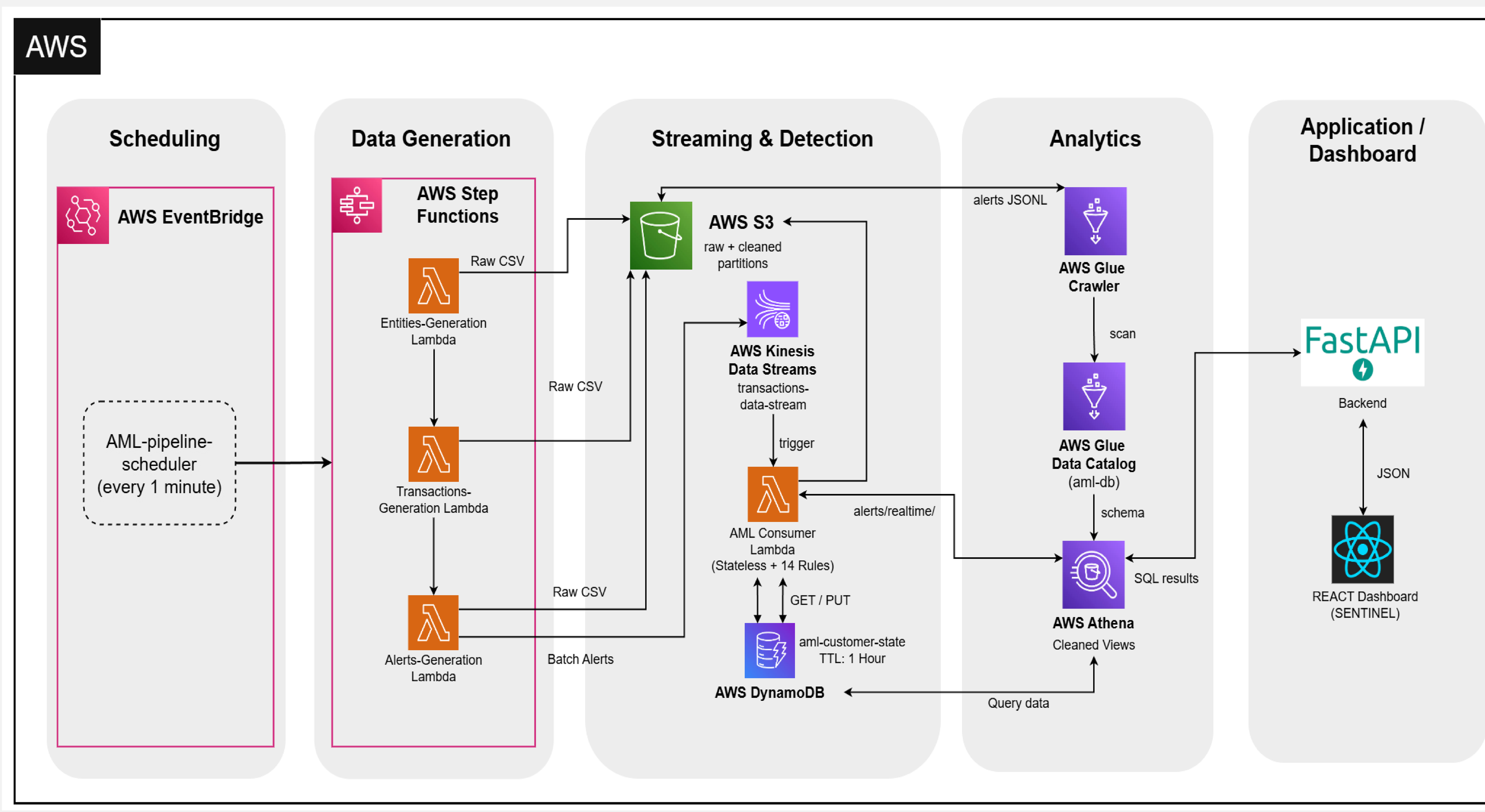
Pattern	Description
HIGH_VALUE	Transaction exceeding a set threshold
STRUCTURING	Breaking a large amount into smaller transactions
SANCTIONED_COUNTRY	Nations under official government or UN sanctions
STRUCTURING_WINDOW	Account's cumulative total across a rolling time window
LAYERING	Funds are bounced through multiple accounts to obscure their origin
VELOCITY_BURST	Automated or coordinated behaviors rather than a natural human pattern
RAPID_MOVEMENT	Funds arriving in an account and leaving again almost immediately
LARGE_RAPID	Combines the severity of HIGH_VALUE with RAPID_MOVEMENT
ROUND_TRIP	Money that ultimately returns to the account it originated from
IMPOSSIBLE_TRAVEL	Same account appears to initiate transactions from two geographically distant locations
SHELL_COMPANY	Classic shell company behavior
HIGH_RISK_COUNTRY	Countries where dirty money is easier to clean
CROSS_BORDER_HIGH	International transfer + a large amount
TRADE_BASED	Transactions disguised as legitimate trade payments

4. DASHBOARD PRIVEWS



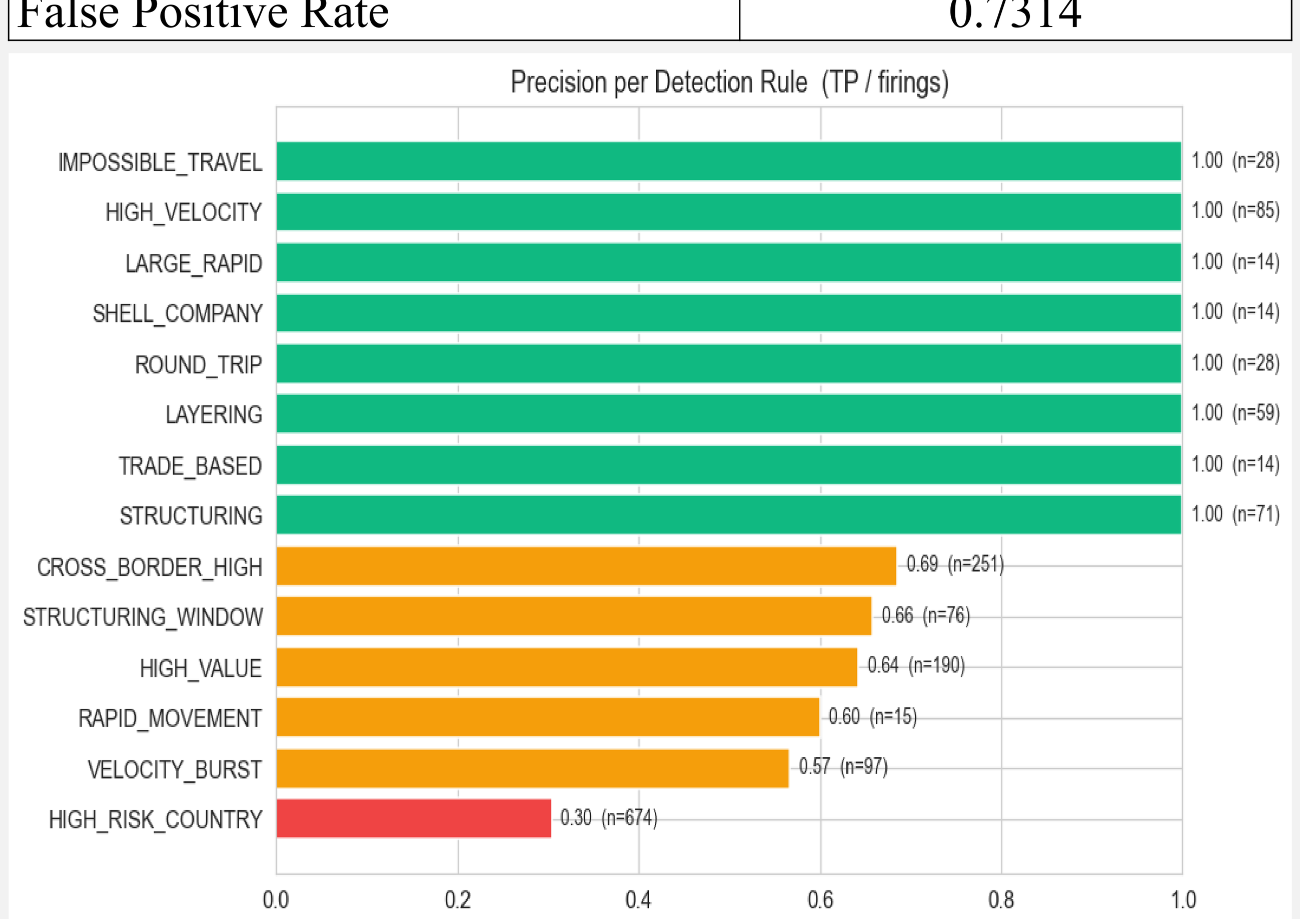


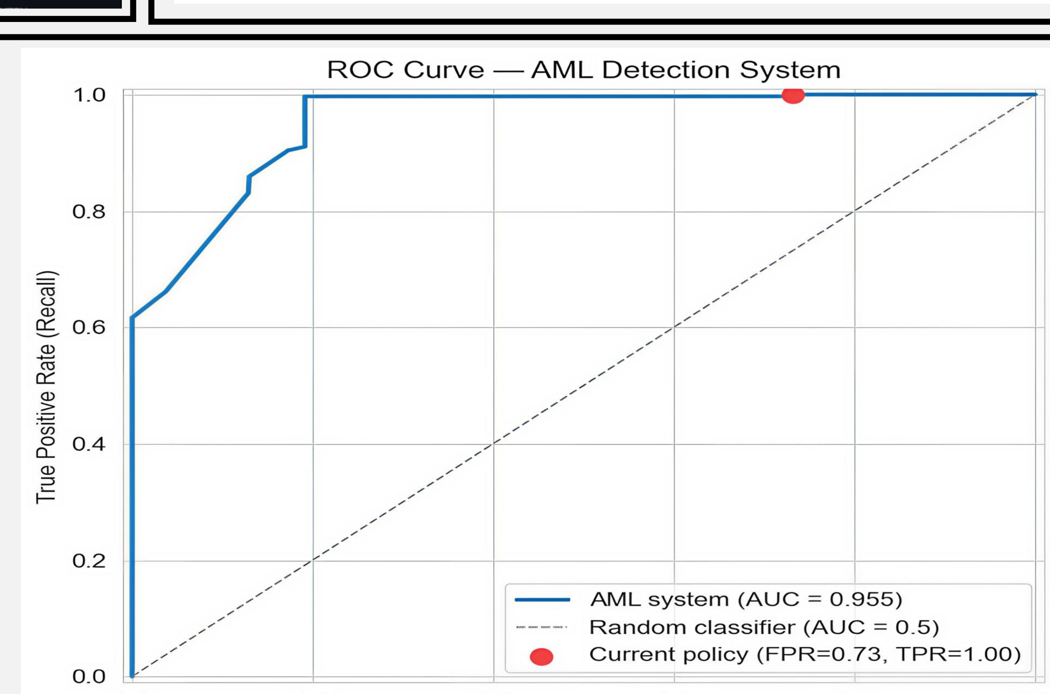
3. SYSTEM ARCHITECTURE

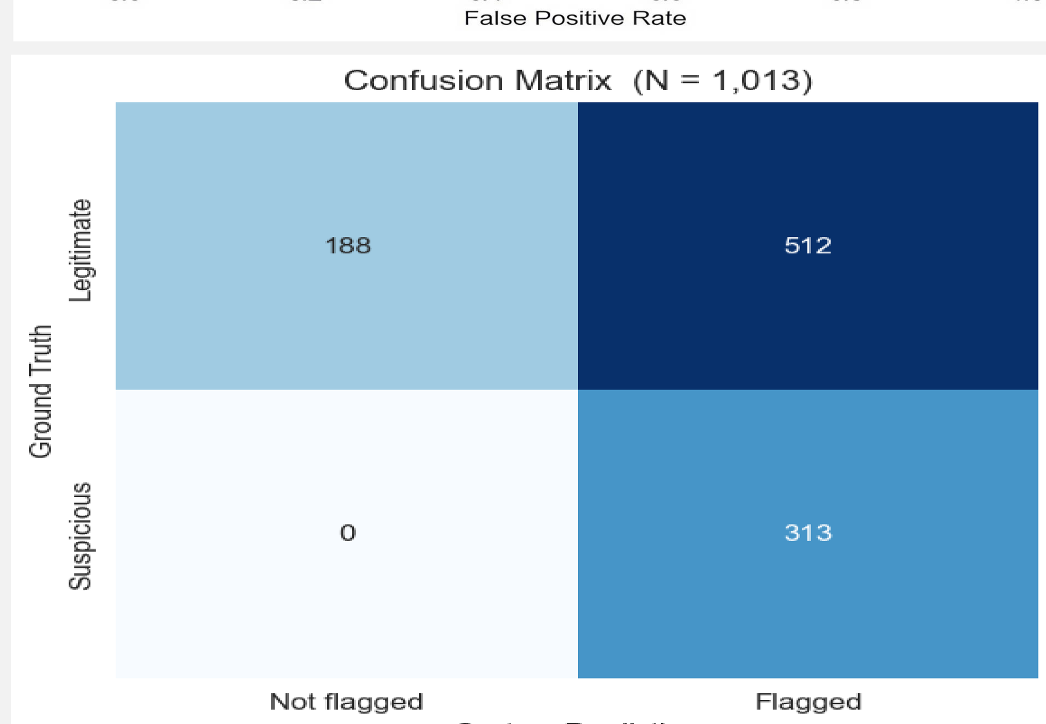


5. RESULTS & PERFORMANCE

Metric	Value
Precision	0.3794
Recall (True Positive Rate)	1.0000
F1 score	0.5501
Accuracy	0.4946
False Positive Rate	0.7314







6. TECHNOLOGIES USED

Cloud & Infrastructure

AWS, Lambda, EventBridge, Glue, S3, StepFunctions, Athena, DynamoDB, Kinesis

Application Stack

REACT, Fast API, TypeScript, Python

Tools & Services

AWS IAM, CloudWatch, Git & GitHubP, VS Code

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